

DETAILED DESCRIPTION OF THE INVENTION — MULTI-SESSION LEARNING, ADAPTIVE CALIBRATION, AND CONTINUOUS IMPROVEMENT SYSTEMS

The intraluminal physiologic monitoring device may employ multi-session learning and adaptive calibration systems configured to improve measurement accuracy and interpretive capability over repeated use. Continuous improvement frameworks may allow the device and associated computational systems to evolve alongside individual physiologic patterns.

In various embodiments, calibration parameters established during initial device use may be refined across subsequent measurement sessions. Adaptive calibration may compensate for anatomical variability, insertion differences, environmental factors, or sensor drift occurring over time.

Multi-session learning systems may aggregate historical physiologic data to identify recurring patterns, long-term trends, and individualized behavioral signatures. Machine learning models may update baseline parameters dynamically based on accumulated longitudinal information.

Adaptive algorithms may adjust sensing sensitivity, sampling frequency, noise filtering thresholds, or signal interpretation rules to maintain measurement reliability under varying conditions. Continuous learning may therefore reduce dependence on fixed calibration assumptions.

Certain embodiments may implement confidence scoring mechanisms that quantify reliability of measurements based on historical performance, signal stability, or contextual consistency. Confidence metrics may guide interpretation weighting or user feedback.

Continuous improvement systems may operate locally on-device, within companion applications, or through distributed cloud-based computational environments. Learning updates may occur in real time or during scheduled synchronization periods.

Population-level learning may be incorporated using anonymized aggregated datasets enabling refinement of predictive models without exposing individual user information. Federated learning techniques may allow collaborative model training while preserving privacy.

The adaptive learning and continuous improvement frameworks described herein may enhance personalization, predictive accuracy, and long-term usability while maintaining privacy-preserving data handling practices.

The multi-session learning and adaptive calibration concepts described are illustrative rather than limiting. Any method enabling improvement of physiologic monitoring performance through accumulated usage data is considered within the scope of the present disclosure.